



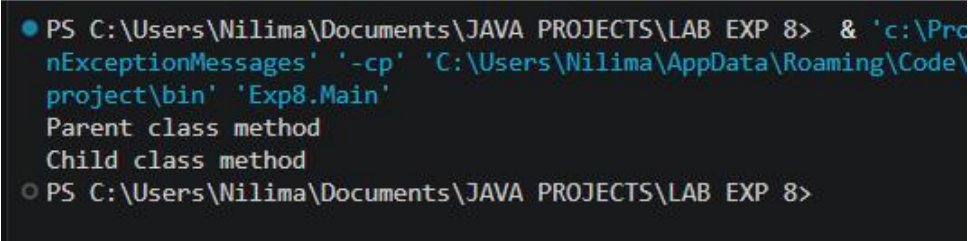
Symbiosis Institute of Technology, Pune

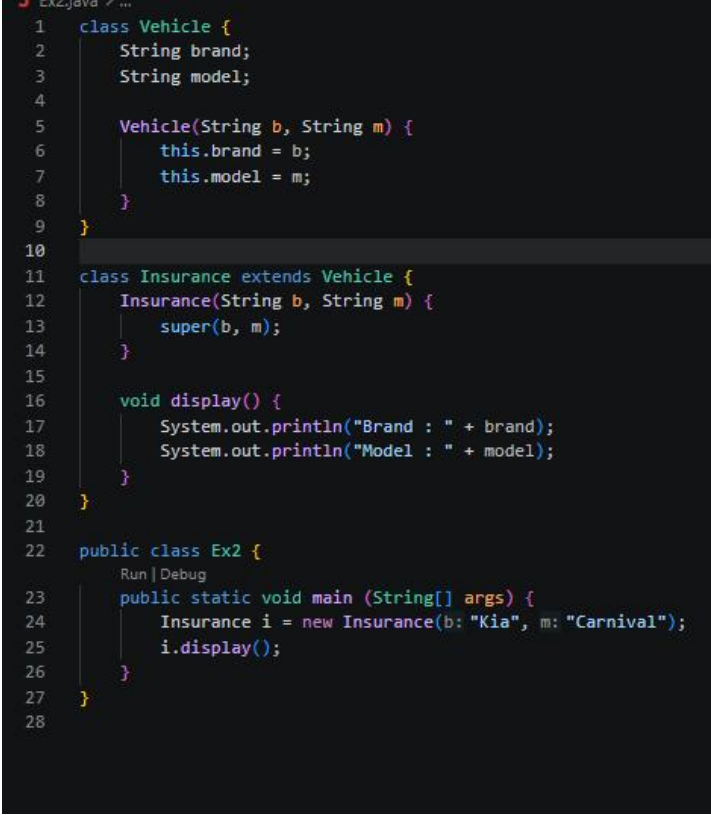
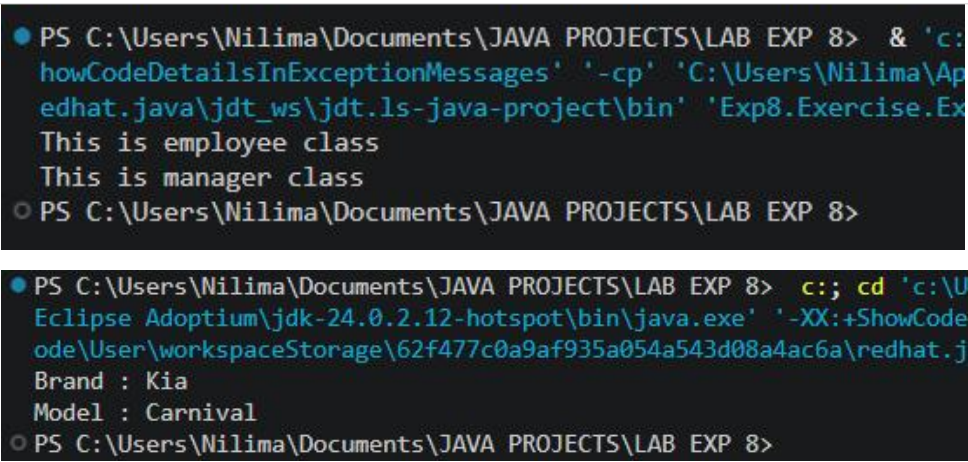
CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No.: 8

Name of Student	Akshit Gawade
PRN No.	25070122029
Batch	2025-2029
Class	A1
Academic Year & Semester	Second Year Semester III
Date of Submission	11/7/2026
Title of Assignment:	Write a Java program to demonstrate the use of the super keyword in inheritance.
Problem Statement	Write a Java program to demonstrate the use of the super keyword in inheritance.
Learning Objectives	To demonstrate the use of the super keyword for accessing parent class members
Theory	The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. It ensures proper constructor chaining during object creation.

Program Code	<pre> 1 class Parent { 2 void display () { 3 System.out.println("Parent class method"); 4 } 5 } 6 7 class Child extends Parent { 8 void display() { 9 super.display(); 10 System.out.println("Child class method"); 11 } 12 } 13 14 public class Main { 15 public static void main (String[] args) { 16 Child c = new Child(); 17 c.display(); 18 } 19 } 20 </pre>
Output Screenshots	 <pre> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 8> & 'c:\Program Files\Java\jdk-9.0.4\bin\java.exe' -cp 'C:\Users\Nilima\AppData\Roaming\Code\project\bin' 'Exp8.Main' Parent class method Child class method PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 8> </pre>
Exercise Qns	Q1. Create an Employee and Manager program where Manager accesses Employee details using super keyword. <pre> 1 class Employee { 2 void display() { 3 System.out.println(x: "This is employee class"); 4 } 5 } 6 7 class Manager extends Employee { 8 void display() { 9 super.display(); 10 System.out.println(x: "This is manager class"); 11 } 12 } 13 14 public class Ex1 { 15 Run Debug 16 public static void main (String[] args) { 17 Manager m = new Manager(); 18 m.display(); 19 } 20 } </pre> Q2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

	 <pre> 1 class Vehicle { 2 String brand; 3 String model; 4 5 Vehicle(String b, String m) { 6 this.brand = b; 7 this.model = m; 8 } 9 } 10 11 class Insurance extends Vehicle { 12 Insurance(String b, String m) { 13 super(b, m); 14 } 15 16 void display() { 17 System.out.println("Brand : " + brand); 18 System.out.println("Model : " + model); 19 } 20 } 21 22 public class Ex2 { 23 public static void main (String[] args) { 24 Insurance i = new Insurance(b: "Kia", m: "Carnival"); 25 i.display(); 26 } 27 } 28 </pre>
Output Screenshots	 <pre> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 8> & 'c:\Program Files\Eclipse Adoptium\jdk-24.0.2-hotspot\bin\java.exe' '-cp' 'C:\Users\Nilima\AppData\Local\Programs\Eclipse Adoptium\jdk-24.0.2-hotspot\bin\java.exe' 'Exp8.Exercise.Ex' This is employee class This is manager class PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 8> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 8> c:; cd 'c:\Program Files\Eclipse Adoptium\jdk-24.0.2-hotspot\bin\java.exe' '-XX:+ShowCodeDetails' '-cp' 'C:\Users\Nilima\AppData\Local\Programs\Eclipse Adoptium\jdk-24.0.2-hotspot\bin\java.exe' 'Exp8.Exercise.Ex' Brand : Kia Model : Carnival PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 8> </pre>
Conclusion	This assignment helped in understanding the use of the super keyword in Java inheritance. It demonstrated how the super keyword is used to access parent class constructors, variables, and methods from a child class, ensuring proper initialization and reducing code duplication. The implementation improved my understanding of inheritance, constructor chaining, and method overriding in object-oriented programming. These concepts are widely used in developing desktop applications, web applications, Android applications, enterprise software, and other real-world Java programs that require efficient code reuse, maintainability, and hierarchical class design.